

SALEKHARD, Russian Federation

WMO: 233300

Lat: 66.5319N

Lon: 66.6744E

Elev: 53

StdP: 14.67

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.7	-38.0	-45.7	0.4	-41.8	-41.6	0.5	-37.6	21.2	-6.4	18.6	-5.1	2.8	180	0.788

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.2	77.7	65.4	73.0	62.7	69.0	60.3	66.6	75.2	64.0	71.3	61.5	68.0	6.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.4	87.5	71.0	60.8	79.6	68.4	58.2	72.4	65.6	31.4	75.0	29.3	71.2	27.5	68.2	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.8	16.8	14.5	DB	-44.6	82.9	6.7	3.7	-49.4	85.5	-53.4	87.6	-57.2	89.7	-62.0	92.4	(4)
(5)			WB	-44.3	70.7	6.1	3.2	-48.7	73.0	-52.3	74.9	-55.7	76.7	-60.2	79.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.4	-10.3	-8.8	5.4	18.0	31.8	50.2	58.0	52.4	41.8	26.6	5.1	-4.0
	DBStd	26.79	16.19	16.17	14.31	12.38	10.21	9.31	7.78	6.55	6.82	10.71	15.79	16.80
	HDD50	10604	1869	1647	1382	959	572	111	14	48	259	724	1346	1673
	HDD65	15598	2334	2067	1847	1409	1030	451	247	395	695	1189	1796	2138
	CDD50	520	0	0	0	0	6	118	261	121	14	0	0	0
	CDD65	40	0	0	0	0	0	8	29	3	0	0	0	0
	CDH74	309	0	0	0	0	0	69	221	19	0	0	0	0
	CDH80	51	0	0	0	0	0	9	41	1	0	0	0	0
Wind	WSAvg	6.2	5.2	5.3	6.0	7.1	7.3	7.7	7.0	6.2	6.1	6.4	5.5	5.3
Precipitation	PrecAvg	17.6	0.9	0.7	0.8	1.0	1.4	2.2	2.5	2.4	1.9	1.7	1.1	1.0
	PrecMax	25.6	1.8	1.7	1.8	3.1	3.2	4.9	7.1	6.3	5.2	3.0	2.3	1.9
	PrecMin	10.9	0.3	0.1	0.0	0.1	0.4	0.5	0.2	0.7	0.3	0.2	0.3	0.0
	PrecStd	3.2	0.4	0.4	0.4	0.6	0.7	1.1	1.5	1.3	0.9	0.6	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.5	30.0	36.6	44.8	66.2	80.8	84.3	76.4	65.6	51.0	35.6	31.7
		MCWB	27.0	28.5	33.4	39.4	55.3	63.6	68.1	65.2	59.0	47.7	34.1	29.7
	2%	DB	23.4	24.2	33.2	40.0	57.9	75.0	79.9	71.0	59.3	46.5	33.0	27.2
		MCWB	22.1	23.0	30.8	36.3	49.5	61.8	66.7	62.0	53.2	43.4	31.8	26.0
	5%	DB	18.5	20.0	29.9	37.7	50.5	70.2	75.7	67.3	55.3	42.5	30.3	23.2
		MCWB	17.6	19.1	27.9	34.8	44.7	59.1	65.2	59.6	50.9	40.3	29.1	22.3
	10%	DB	13.7	15.2	26.5	35.5	45.2	65.6	71.1	63.8	52.1	39.4	26.2	18.9
		MCWB	13.0	14.3	25.0	33.2	40.7	56.7	62.4	57.9	48.5	37.6	25.1	17.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	28.4	33.8	40.3	55.5	66.1	71.6	67.5	59.4	48.3	34.2	29.8
		MCDB	28.4	29.9	35.8	43.8	65.0	76.4	79.7	74.5	64.6	50.4	35.2	31.1
	2%	WB	22.3	23.2	31.0	36.9	49.6	63.2	68.5	63.1	54.5	43.6	32.1	26.1
		MCDB	23.4	24.2	33.0	39.2	57.1	73.3	77.3	68.5	58.0	46.1	32.9	27.1
	5%	WB	17.6	19.2	28.0	35.1	45.3	60.4	65.9	60.8	51.6	40.3	29.2	22.3
		MCDB	18.4	20.0	29.7	37.3	50.4	69.1	74.1	66.1	54.5	42.2	30.3	23.2
	10%	WB	13.0	14.3	25.0	33.3	41.1	57.3	63.5	58.4	48.8	37.8	25.2	18.0
		MCDB	13.8	15.2	26.5	35.4	44.8	64.3	70.3	63.2	52.0	39.3	26.2	18.7
Mean Daily Temperature Range	5% DB	MDBR	13.5	13.8	16.0	15.6	12.4	14.2	15.2	13.9	11.0	9.4	12.5	13.3
		MCDBR	16.8	15.1	14.7	13.0	19.5	21.3	20.6	19.8	16.1	10.1	12.0	14.5
		MCWBR	16.3	14.5	13.2	10.4	12.6	11.1	10.3	11.9	10.9	8.1	11.5	14.1
	5% WB	MCDBR	16.5	15.2	14.5	12.2	19.4	20.1	19.3	17.3	14.7	9.7	11.9	14.4
		MCWBR	16.0	14.6	13.1	10.3	12.8	11.6	10.7	11.3	10.8	8.0	11.6	14.0
Clear-Sky Solar Irradiance	taub		0.204	0.258	0.307	0.331	0.349	0.363	0.390	0.369	0.320	0.276	0.198	0.210
	taud		1.886	2.042	2.055	2.105	2.252	2.438	2.363	2.392	2.489	2.231	1.920	1.897
	Ebn at Noon		57	181	230	259	268	268	254	248	237	177	56	3
	Edh at Noon		5	18	30	37	36	31	32	28	20	15	4	1
All-Sky Solar Radiation	RadAvg		20	202	712	1383	1741	1657	1657	1070	577	198	42	2
	RadStd		2	20	47	76	127	120	173	88	49	30	4	0

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	+2.77	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page